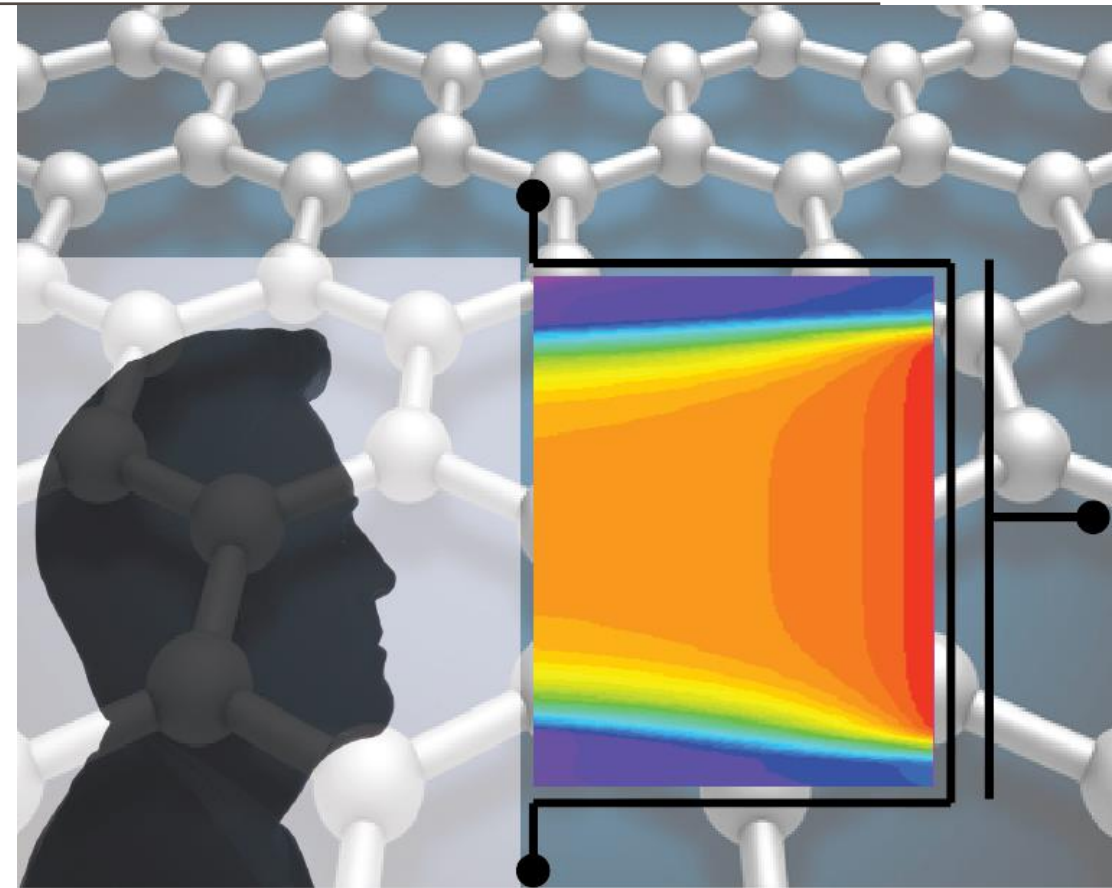


# ADVANCED SOLID-STATE DEVICES

## Week-6, Lecture-1

Sneh Saurabh  
10<sup>th</sup> February, 2025



# Diffusion Resistance: Physical Meaning

- Assume that a diode is biased at a DC voltage  $V_0$  in the forward mode.
- Let us represent the corresponding DC signal as  $I_{DQ}$

- Assume that a small low-frequency AC voltage is superimposed over the DC signal.
- It will produce an AC current superimposed over  $I_{DQ}$
- The ratio of AC current to the AC voltage is the incremental conductance.

- When amplitude of AC signal is infinitesimally small, we can write incremental conductance as :

$$g_d = \left. \frac{dI_D}{dV_a} \right|_{V_a=V_0}$$

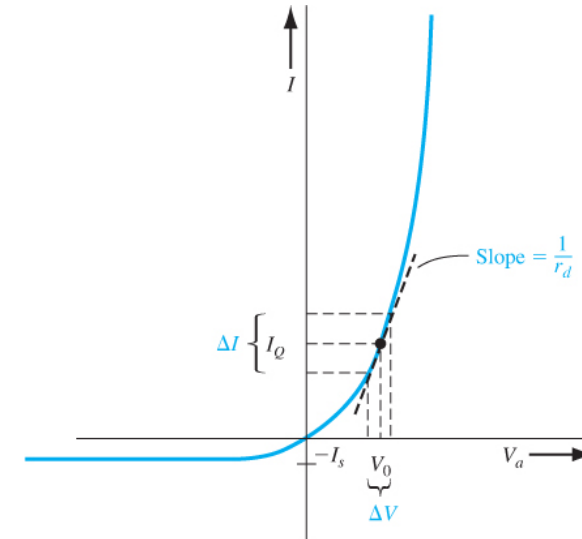


Figure 8.18 | Curve showing the concept of the small-signal diffusion resistance.

- The reciprocal of incremental conductance is incremental resistance (also called **diffusion resistance**)

$$r_d = \left. \frac{dV_a}{dI_D} \right|_{I_D=I_{DQ}}$$

# Diffusion Resistance: Computation

**Problem:** Assume that a diode current is given as:

$$I_D = I_s \left[ \exp \left( \frac{qV_a}{kT} \right) - 1 \right]$$

Compute the diffusion resistance at a given  $I_{DQ}$

$$r_d = \left. \frac{dV_a}{dI_D} \right|_{I_D=I_{DQ}}$$

**Answer:**  $r_d = \frac{kT/q}{I_{DQ}}$

- Diffusion resistance *decreases* as  $I_{DQ}$  increases

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PN Junction

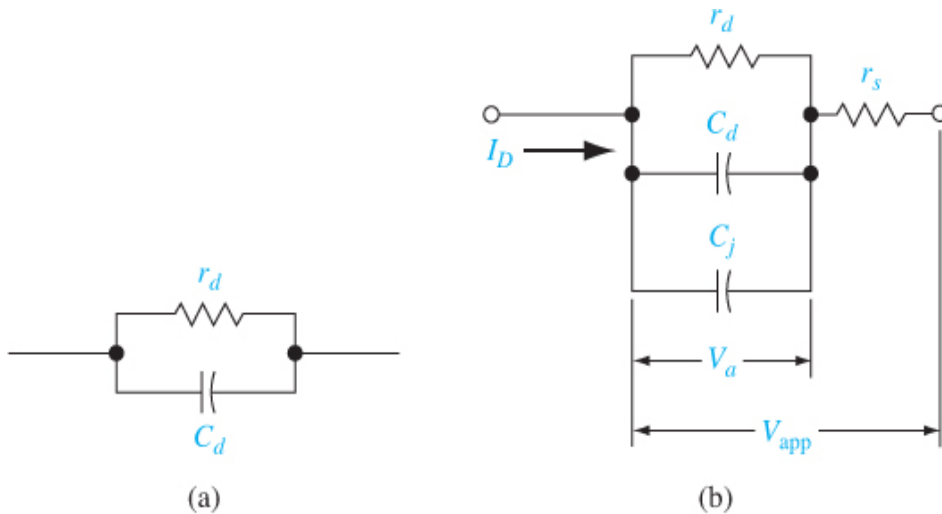


Small-signal Model

# Small Signal Equivalent Circuit

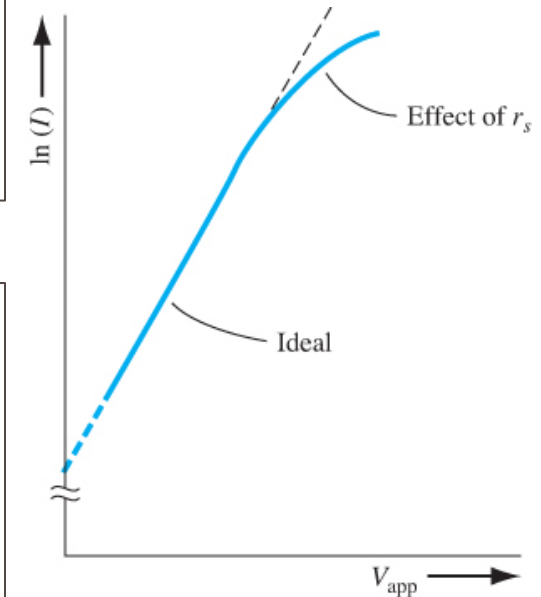
- In forward-biased mode: diffusion capacitance and resistance are in parallel

- We need to add junction capacitance also in parallel [Will be dominant in reverse mode]



**Figure 8.22** | (a) Small-signal equivalent circuit of ideal forward-biased pn junction diode; (b) complete small-signal equivalent circuit of pn junction.

- A series resistance is added because the neutral p/n regions will have some finite resistance



**Figure 8.23** | Forward-biased  $I$ - $V$  characteristics of a pn junction diode showing the effect of series resistance.

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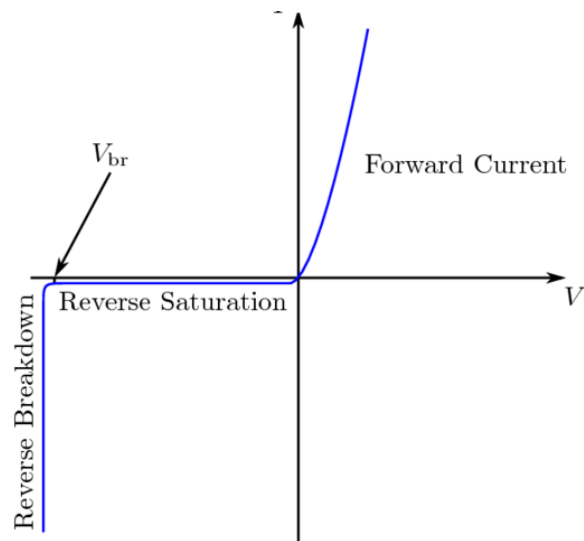
PN Junction



Breakdown

# Junction Breakdown: Meaning

- In reverse-biased mode, reverse current increases only slightly with the bias increase.
- However, when a **critical voltage** is reached, the reverse current increases abruptly with the applied bias
- This phenomenon is the **breakdown of the pn junction**



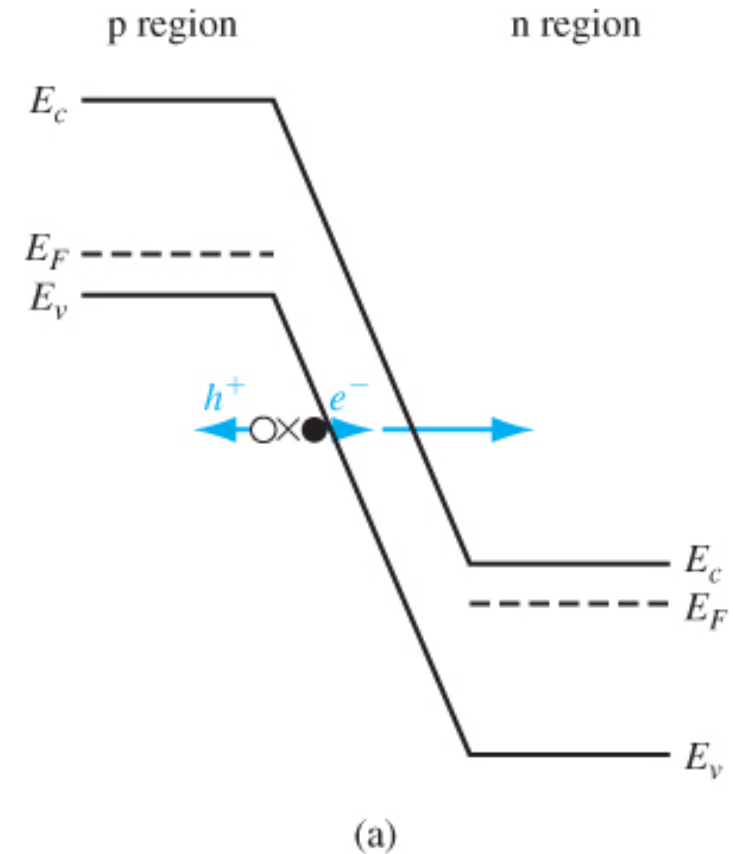
- Breakdown is electrical, there is no mechanical damage to the device
- The breakdown is reversible and non-destructive (if the current meets thermal consideration)

Two types:

1. **Zener breakdown:** occurs in highly doped PN junction, depletion region width small and tunnelling is highly probable
2. **Avalanche:** due to secondary multiplication, occurs in junctions with thicker depletion regions

# Zener Breakdown

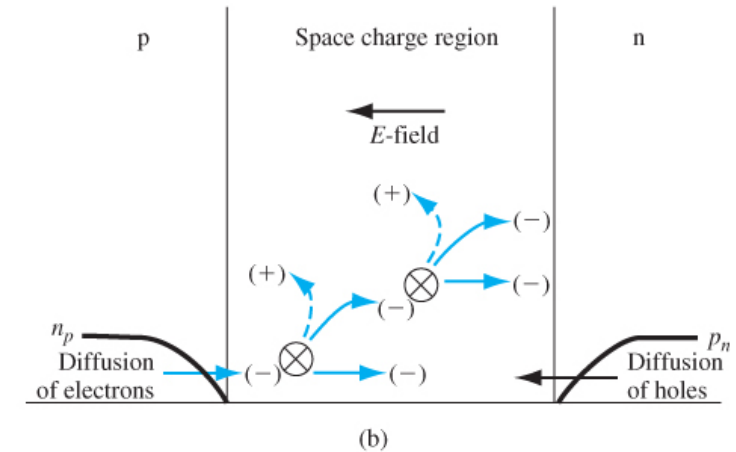
- In highly doped PN junction, the conduction band on the n-side is very close to the valence band on the p-side
- When sufficient reverse bias voltage is applied, the bands align and the width of barrier decreases
  - Electrons can tunnel from valence band on the p-side to the conduction band on the n-side
  - High current can be obtained in reverse bias also





# Avalanche Breakdown

- Electrons/holes when move through depletion region gain energy from the electric field
- When the gained energy is sufficient, they will create electron-hole pairs from the atoms in the depletion region
- Newly created electron-hole pairs move in opposite direction due to electric field: contribute to the reverse current



- The newly created electrons further gain high energies and **more electron-hole pairs are created**
- The electron concentration consequently *multiplies*
- This is known as *avalanche multiplication* and has a **positive feedback**
- Eventually the junction or transition region breaks down and the current rises sharply

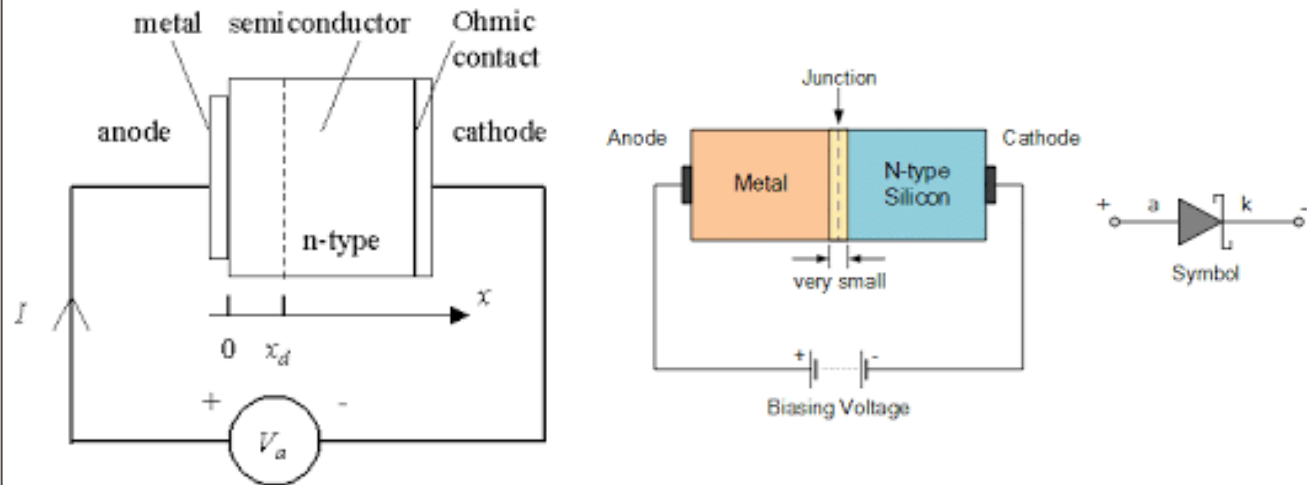
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# Metal—Semiconductor Junction

# Metal—Semiconductor Junction (1)

- Junction formed when metal and semiconductor are in contact
- Examples:
  - Metal contacts
  - Schottky Diodes



- The behaviour of both metal and semiconductor can vary
  - Metal can be attracting electrons more/less
  - Semiconductor can be n/p-type and high/low doping, band gap

Can show two types of behaviour:

1. **Rectifying:** current flows easily in one direction only
2. **Non-rectifying:** current flows easily in both directions

We will understand how do these behaviours originate

# Metal—Semiconductor Junction (2)

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What properties of metal and semiconductor determine the characteristics of M-S junctions?

Material properties that are most important in determining the characteristics of M-S junction

**1. Metal:**

- Work function

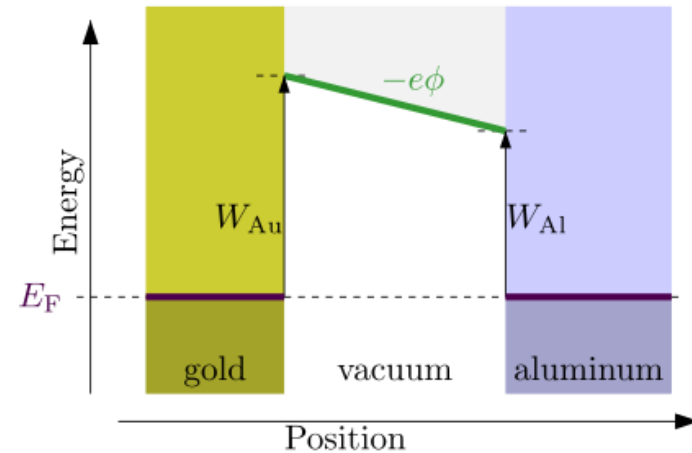
**2. Semiconductor:**

- Work function
- Electron affinity

# Work function

- What is *work function*?
  - Energy required to bring an electron from the *Fermi level* to the *vacuum level*
  - Vacuum level represents the energy of an electron at rest outside the solid (metal or semiconductor)

- We denote work function as  $\phi_m$  and  $\phi_s$  (unit is typically eV)



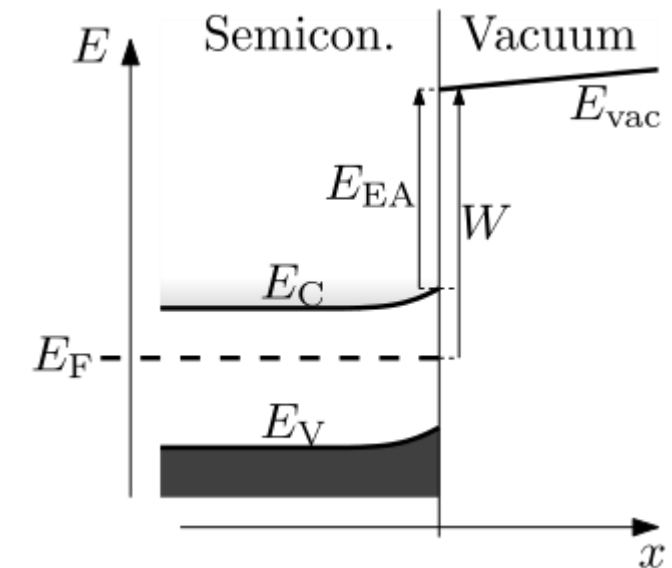
**Table 9.1** | Work functions of some elements

| Element        | Work function, $\phi_m$ |
|----------------|-------------------------|
| Ag, silver     | 4.26                    |
| Al, aluminum   | 4.28                    |
| Au, gold       | 5.1                     |
| Cr, chromium   | 4.5                     |
| Mo, molybdenum | 4.6                     |
| Ni, nickel     | 5.15                    |
| Pd, palladium  | 5.12                    |
| Pt, platinum   | 5.65                    |
| Ti, titanium   | 4.33                    |
| W, tungsten    | 4.55                    |

# Electron Affinity

- What is electron affinity?
  - The energy obtained by moving an electron from the **vacuum level** just outside the semiconductor to the bottom of the **conduction band** just inside the semiconductor

- We denote work function as  $E_{EA}$  or  $\chi$



**Table 9.2** | Electron affinity of some semiconductors

| Element                 | Electron affinity, $\chi$ |
|-------------------------|---------------------------|
| Ge, germanium           | 4.13                      |
| Si, silicon             | 4.01                      |
| GaAs, gallium arsenide  | 4.07                      |
| AlAs, aluminum arsenide | 3.5                       |

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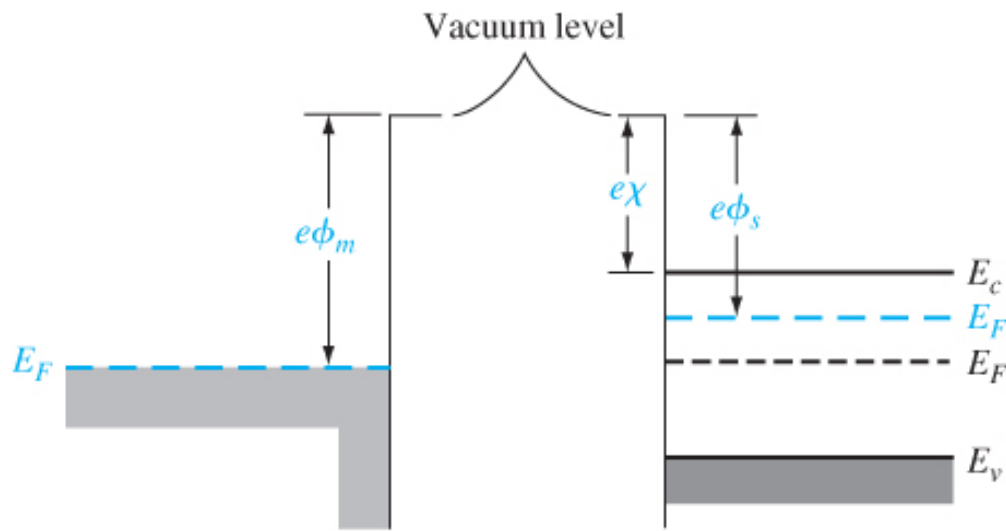
# Metal—Semiconductor Junction



## Rectifying

# Rectifying MS Junction: Equilibrium (1)

**Problem:** The band diagram of metal and n-type semiconductor when they are far apart is shown below. Assume  $\phi_m > \phi_s$ . Draw the band diagram when they are in contact.



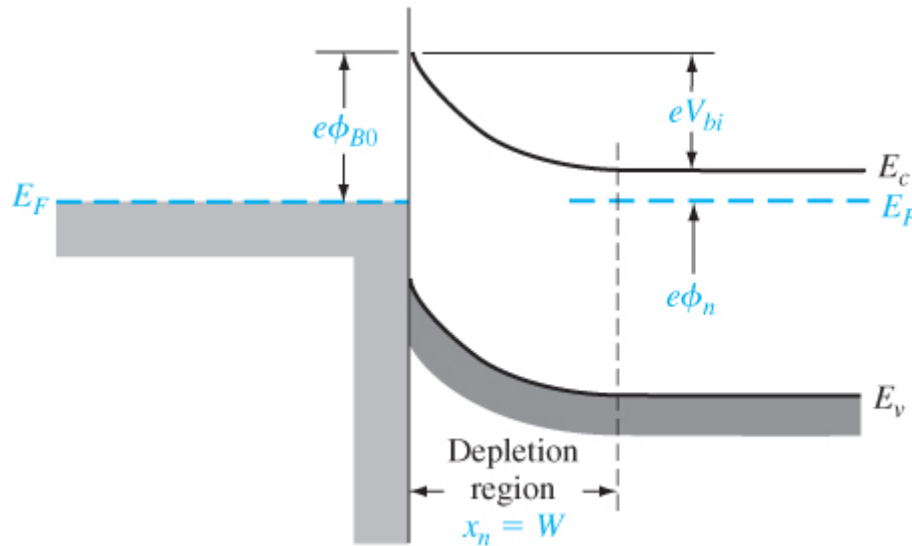
- At equilibrium, Fermi energy will be same throughout the system

- To align Fermi energy, electrons will flow from semiconductor to metal (Why?)
  - Electrons in conduction band are at higher energy

- When electrons move from n-type semiconductor, what will happen to the energy bands of the semiconductor?
  - It will move towards becoming intrinsic
  - More gap will be created between conduction band edge and Fermi energy near the junction
  - The energy bands will bend *upward* near the junction on the semiconductor side



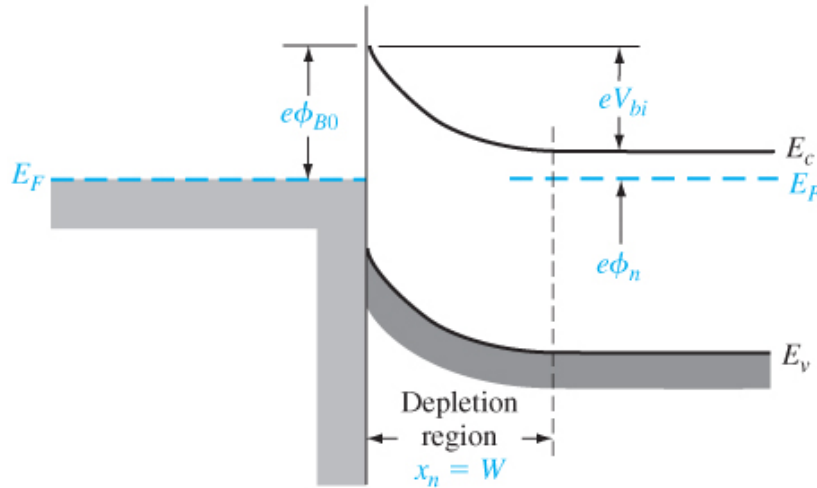
# Rectifying MS Junction: Equilibrium (2)



- When electrons move from n-type semiconductor, what will happen to the charge neutrality in the semiconductor?
  - It will have immobile positive donor ions
  - A depletion region of width  $x_n = W$  will be created

- What will happen to the junction on the metal side?
  - Electrons will be contained within a distance of about an atomic layer
  - Surface negative charge will be accumulated on the metal
- The structure is similar to PN junction
  - Electric field will exist from semiconductor to metal
  - Electrons on the metal side will see a barrier for moving to the semiconductor side
  - Electron in the conduction band (on semiconductor side) will see a barrier for moving to the metal side
  - At equilibrium these barriers obstruct current flow

# Rectifying MS Junction: Two Barriers



At thermal equilibrium (at room temperature), do we expect that: a) no electrons will cross the junction, or b) some electrons will cross the junction?

At room temperature, some electrons will have energy more than the barriers (why?)

- Energy distribution is decided by statistically (FD Distribution)

- Barrier seen on the metal side for electrons:

$$\phi_{B0} = (\phi_m - \chi)$$

- This barrier is known as *Schottky barrier* (ideal barrier height)

- Barrier seen on the semiconductor side for electrons is  $V_{bi}$  similar to PN junction:

$$V_{bi} = (\phi_{B0} - \phi_n)$$

- The built-in potential is a function of doping also

- These high energy electrons will cross the barrier and reach other side

- At equilibrium, current in both directions are equal and the net current is zero

- At non-equilibrium (applying bias) these components lose balance

# Metal-Semiconductor Junction: Problem

Consider a contact between Tungsten and n-type semiconductor with  $N_d = 10^{16} \text{cm}^{-3}$  at 300 K. The work function of tungsten is  $\phi_m = 4.55 \text{ V}$  and the electron affinity of silicon is  $\chi = 4.01 \text{ V}$ .

Determine the theoretical barrier height and built-in potential barrier.

[Take  $\epsilon_s = 11.7 \times 8.85 \times 10^{-14} \text{ F/cm}$  and  $N_c = 2.8 \times 10^{19} \text{cm}^{-3}$ ]

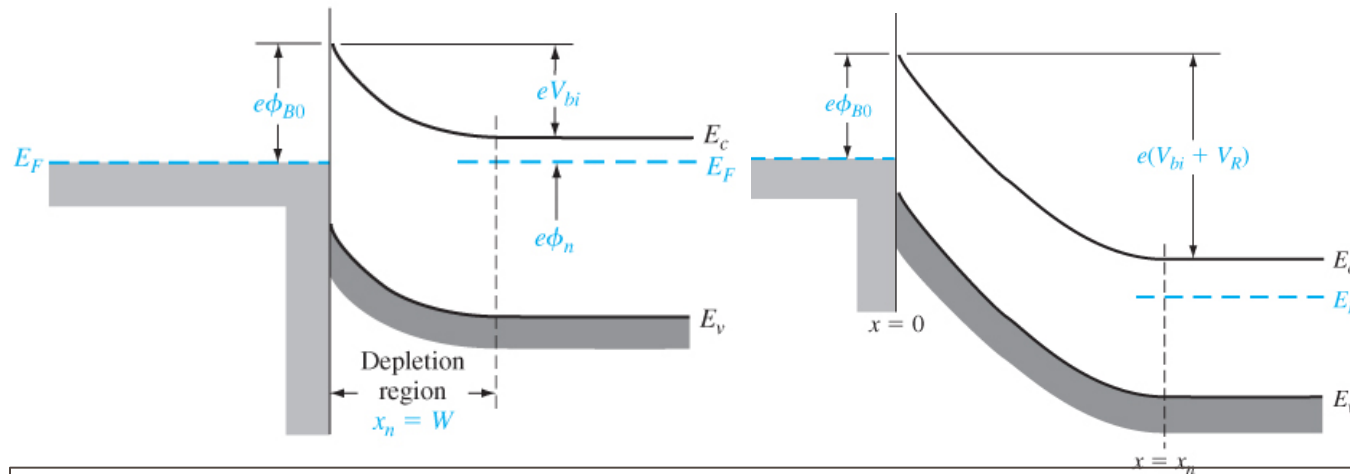
$$\begin{aligned}\phi_{B0} &= (\phi_m - \chi) \\ V_{bi} &= (\phi_{B0} - \phi_n)\end{aligned}$$

$$\begin{aligned}\phi_{B0} &= 0.54 \text{ V} \\ \phi_n &= 0.206 \text{ V} \\ V_{bi} &= 0.334 \text{ V}\end{aligned}$$

$$\phi_n = \frac{kT}{q} \ln \left( \frac{N_c}{N_d} \right)$$

# Rectifying MS Junction: Reverse-biased

**Problem:** Assume that the band diagram of M-S junction in equilibrium is as shown below. Draw the band diagram for the case when we apply *positive voltage*  $V_R$  to the semiconductor with respect to metal.

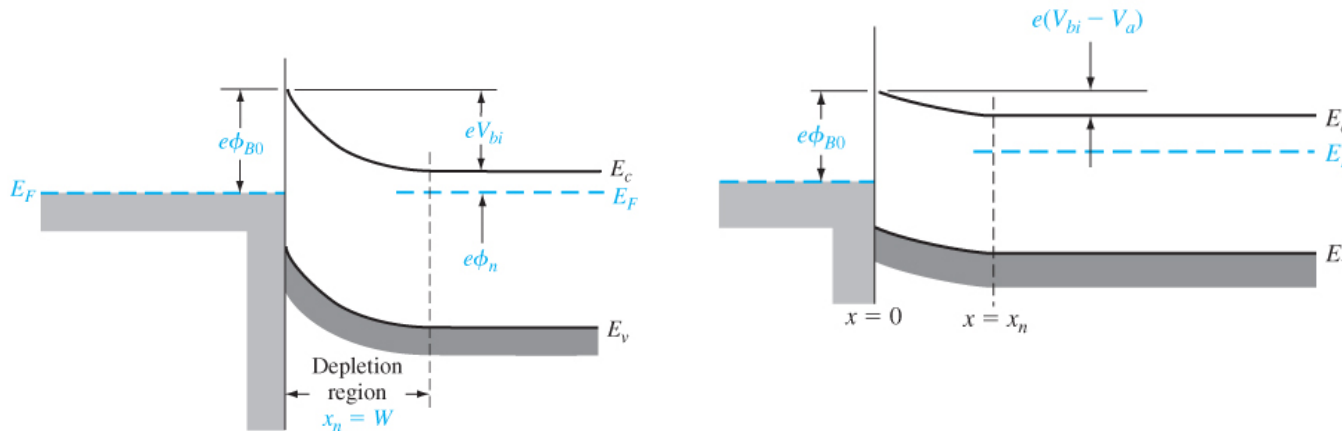


- Electron flow from semiconductor to metal decreases
- Electron flow from metal to semiconductor remains same
- Total: Some electron flow from metal to semiconductor (small conventional current from semiconductor to metal)

- The bands move downwards on the semiconductor side and remains unaffected on the metal side
  - The barrier on the semiconductor side increase to  $V_{bi} + V_R$ , and remains  $\phi_{B0}$  on metal side
- 
- Situation similar to reverse-biased PN junction

# Rectifying MS Junction: Forward-biased

**Problem:** Assume that the band diagram of M-S junction in equilibrium is as shown below. Draw the band diagram for the case when we apply *negative voltage*  $V_a$  to the semiconductor with respect to metal.



- Electron flow from semiconductor to metal increase significantly
- Electron flow from metal to semiconductor remains same
- Total: **Significant** electron flow from semiconductor to metal (conventional current from semiconductor to metal)

- The bands move upward on the semiconductor side and remains unaffected on the metal side
- The barrier on the semiconductor side decrease to  $V_{bi} - V_a$ , and remains  $\phi_{B0}$  on metal side
- Situation similar to forward-biased PN junction
- Thus, MS junction acts as rectifying junction in this case.

# References

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- D. A. Neaman, "*Semiconductor Physics and Devices*"